

Reproducing VISAtlas

An Image-Based Exploration and Query System for Large Visualization Collections

September 24, 2026

“The visualization collections are inherently diverse in many perspectives: from different data sources, by different research groups, and in different manners.”

“Existing systems for visualization images ... are mainly based on extrinsic attributes including author names and publication years.”

“More importantly, attribute-based systems neglect the intrinsic property, i.e., visual appearance of visualizations that is also perceived by humans.”

Although visual appearance is directly perceived by humans, it has often been overlooked as an analyzable property of visualization collections.

Y. Ye, R. Huang, W. Zeng. VISAtlas. IEEE TVCG 30(7):3224–3240, 2024.

“Which visualization collection shall be used?

Are there any similar visualizations in the collection?

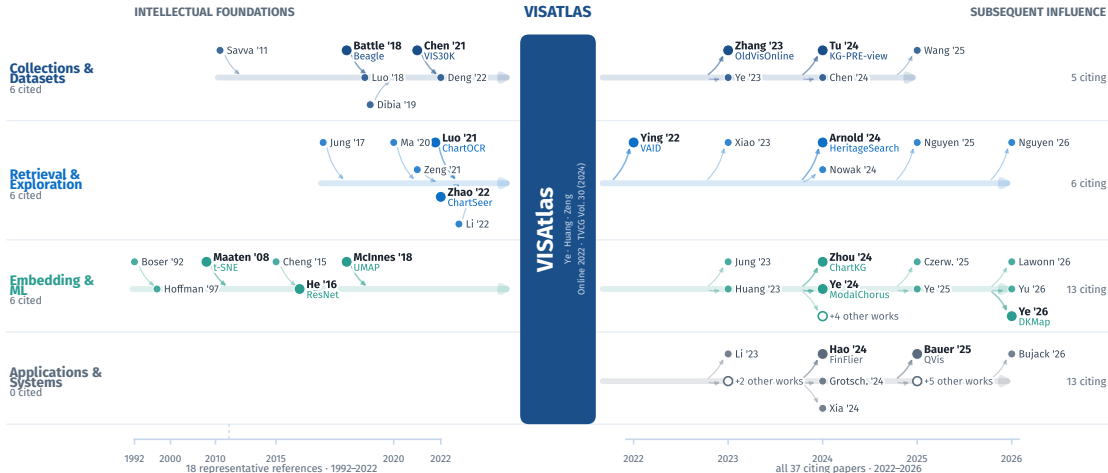
What visual features are prominent in a visualization?

What visual features are frequently composited in visualizations?”

VISAtlas represents the visual appearance of visualizations as interpretable embeddings, enabling users to explore, compare, and query visualization collections based on visual features rather than only extrinsic metadata.

Y. Ye, R. Huang, W. Zeng. VISAtlas. IEEE TVCG 30(7):3224–3240, 2024.

Related Work: Citation Provenance

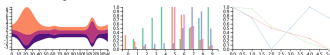


VISAtlas System Overview

What goes in?

Diverse visualization images

Synthetic Visualizations



Diverse chart types, data distributions, and colormaps

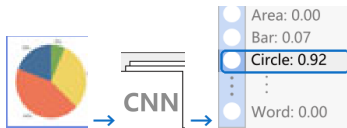
Real-World Visualizations



Web-crawled visualizations reflecting realistic usage

What does VISAtlas do?

Encode visual appearance



Human-interpretable semantic embedding

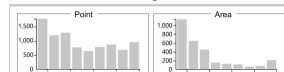
What can users do?

Explore



Collection structure

Compare



Embedding distributions

Query



Similar designs

Diverse visualizations

Interpretable semantic
representation

Reproducing VISAtlas

Visual analytics tasks

Data Collection

Controlled coverage + realistic diversity

Synthetic Visualizations

Controlled coverage



Representative synthetic and demo visualizations from the released VISAtlas resources

Real-World Visualizations

Realistic diversity

Beagle — Web practice



VIS30K — Research practice

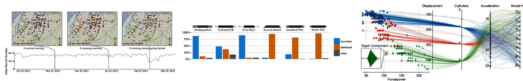


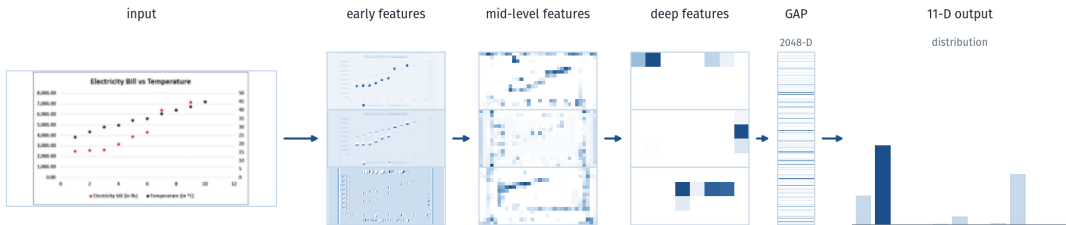
Chart type × Data distribution × Colormap

Synthetic coverage + real-world diversity → a comprehensive visualization collection

Examples from the released VISAtlas resources (github.com/yilinye/VisAtlas-Code)

Image Embedding via CNN

From pixels to feature maps to an 11-D representation



ResNet50 transforms spatial patterns into a compact numerical embedding.

From Triplet Metric Learning to VISAtlas

From identity-based metric learning to visualization-type-aware embeddings

Triplet Metric Learning — FaceNet



$$\|f(A) - f(P)\|_2^2 + \alpha < \|f(A) - f(N)\|_2^2$$

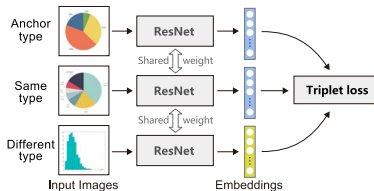
Pull close · Push apart



Triplet learning turns **semantic identity** into **geometric separation**.

Based on FaceNet (Schroff et al., CVPR 2015, Fig. 3); scatter plot is schematic.

VISAtlas: Type Supervision + Triplet Learning



$$L = L_{CE} + \beta L_{TR}$$

$$L_{CE}: q_n \approx p_n$$

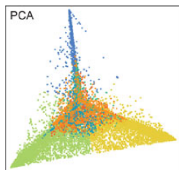
Type semantics

$$L_{TR}: d(q_n, q_m) + \alpha \leq d(q_n, q_l)$$

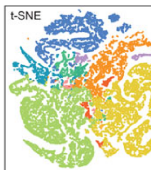
Relative geometry

Why Not Conventional Dimensionality Reduction?

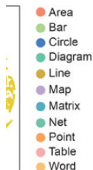
VISAtlas prioritizes semantic interpretability over a generic 2-D projection.



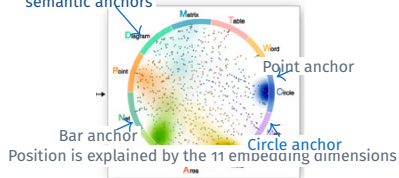
Global structure
× Minor types obscured



Local clusters
× Distances not interpretable



semantic anchors



(d) Sampling + density plot

$$P_n = \sum_i q_n^i v_i$$

- Circle anchor
- ↑ strong weight
- embedding point

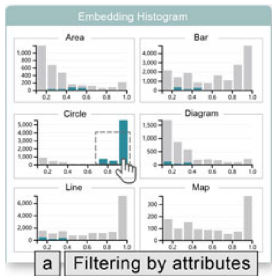
- Bar anchor
- Point anchor

→
generic projection
interpretable
projection

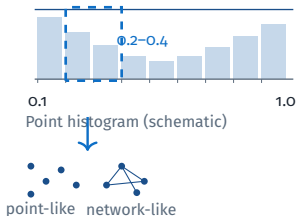
VISAtlas keeps visualization-type semantics visible in the 2-D layout.

Inspecting the 11-D Embedding

Embedding Histogram exposes the distribution of each semantic dimension.



$$q_n = (q_n^1, \dots, q_n^{11}), \quad \boxed{q_n^{\text{Point}}} \rightarrow \text{histogram}$$



- High values → **confident association**
- Broad distribution → **ambiguous type**
- Low-value Point cases → often **Trees & Networks**

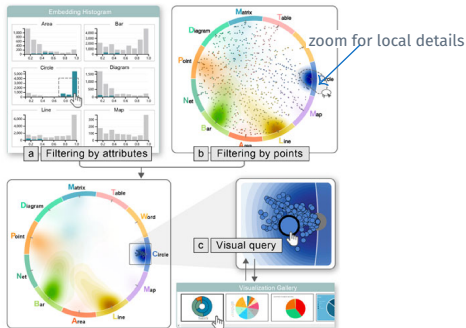
The histogram makes individual embedding dimensions inspectable and exposes model confusion.

Interactive Exploration and Visual Query

Users can filter, select, query, and inspect similar visualizations.

1 FILTER

Histogram bins or
lasso selection



2 QUERY

Select a point or upload an
image

3 RETRIEVE

Return the most similar
visualizations

$q_{\text{query}} \rightarrow$ nearest
embeddings

cosine similarity or projected
Euclidean distance

